UNIT - 5

Vibration Measurement and Applications

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Transducers: variable resistance, piezoelectric, electrodynamic and LVDT transducers; vibration pickups: vibrometer, accelerometer, velometer and phase distortion; frequency-measuring instruments; vibration exciters: mechanical exciters and electrodynamic shaker.

1. Vibration Measurement - Introduction

Vibration measurement is needed for: (i) finding natural frequencies and damping of structures, (ii) checking operating levels against standards, (iii) condition monitoring and fault diagnosis of machines, (iv) verifying analytical models, (v) field balancing, (vi) testing products (qualification, fatigue).

Quantities measured: displacement, velocity, acceleration, frequency, phase. For harmonic motion $x = X \sin \omega t$: $v = \omega X$, $a = \omega^2 X$ (amplitudes).

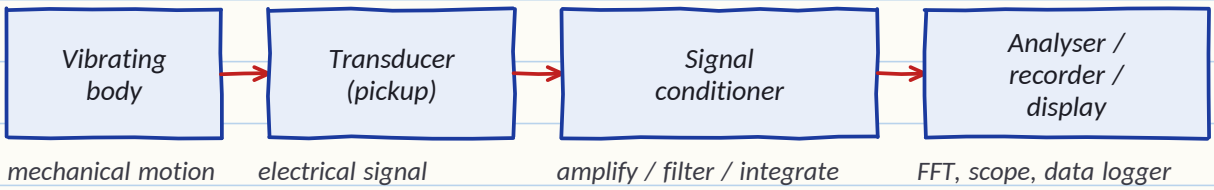


Fig 5.1 Vibration measuring chain

Transducer = device that converts mechanical motion (or force) into an electrical signal (voltage / charge / resistance change).

Classification of transducers

Type	Principle	Examples
Active (self-generating)	Generates its own voltage / charge, needs no external excitation	Piezoelectric, electrodynamic (velocity coil), thermocouple
Passive	Needs external power; motion changes a circuit parameter R, L or C	Strain gauge (R), LVDT (L), capacitive (C), potentiometer (R)

2. Transducers

2.1 Variable resistance transducers (strain gauge)

Electrical resistance of a wire $R = \rho l/A$ changes when it is strained (length and area change, resistivity changes). Gauge factor:

$$GF = (\Delta R/R) / \varepsilon = (\Delta R/R)/(\Delta L/L) \quad (1)$$

metal foil gauge: GF ≈ 2; semiconductor: 50-200

- Bonded foil gauge: fine grid of constantan/nichrome foil on a plastic backing, cemented to the member; measures surface strain (hence stress, force, torque, pressure, acceleration when mounted on a cantilever).
- Resistance change is tiny → measured with a Wheatstone bridge. For quarter bridge (one active gauge) output $e_0 = (E/4)\, GF\, \varepsilon$.
- Also potentiometer (sliding contact on resistor) for low-frequency displacement; unbonded wire gauge accelerometers.
- Advantages: cheap, small, accurate, static and dynamic response (to ~ 10 kHz). Limits: temperature effects (use dummy gauge), low output, fatigue, fragile.

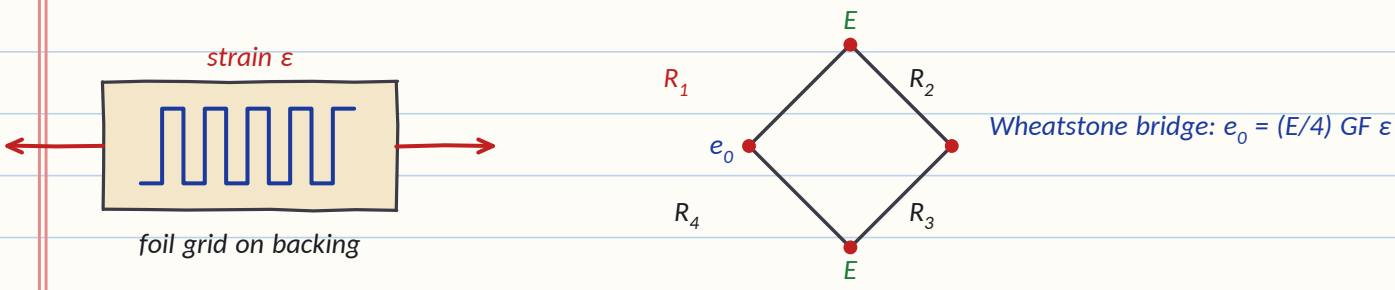


Fig 5.2 Foil strain gauge and bridge

Problem 1 (strain gauge)

A 120 Ω gauge, GF = 2.0, strain 500 με (500×10^{-6}). Find ΔR and quarter-bridge output for E = 10 V.

$\Delta R = R \cdot GF \cdot \varepsilon = 120 \times 2 \times 500\times10^{-6} = 0.12\ \Omega$; $e_0 = (E/4)GF\,\varepsilon = (10/4)(2)(500\times10^{-6}) = 2.5$ mV.

2.2 Piezoelectric transducers

Certain crystals (quartz, Rochelle salt, barium titanate, PZT ceramics) develop an electric charge when mechanically stressed (piezoelectric effect).

$q = d\,F$ (d = charge sensitivity, pC/N; quartz ≈ 2.3 pC/N)

$e_0 = q / C$ (open-circuit voltage; C = crystal + cable + amplifier capacitance)

- Piezo accelerometer: crystal(s) preloaded with a seismic mass; the base acceleration gives force $F = m\,a$ on the crystal → charge ~ acceleration (Fig 5.3).
- Needs a charge amplifier (or ICP built-in amplifier) because of very high output impedance; output affected by cable length if voltage amplifier is used.
- Advantages: very wide frequency range (≈ 1 Hz - 30 kHz), small size, light, rugged, high natural frequency (> 30 kHz), no power (self-generating), good linearity.
- Limits: cannot measure static (DC) acceleration (charge leaks away); temperature and cable noise (triboelectric); high cost.

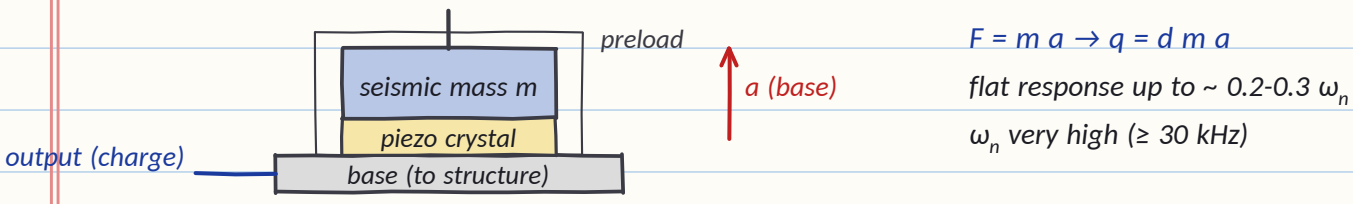


Fig 5.3 Piezoelectric accelerometer (compression type)

Problem 2 (piezo voltage)

Quartz crystal d = 2.3 pC/N loaded by a force amplitude 10 N; total capacitance 100 pF. Find charge and voltage.

$q = dF = 2.3 \times 10 = 23$ pC; $e_0 = q/C = 23\times10^{-12}/100\times10^{-12} = 0.23$ V.

2.3 Electrodynamic (electromagnetic) transducers

A coil moves in a magnetic field of a permanent magnet (or a magnet moves inside a coil). By Faraday's law the induced emf is proportional to the velocity:

$$e = B\,l\,v \quad (3)$$

B (T) flux density, l (m) conductor length in field, v (m/s) relative velocity

- Self-generating (active) - no power supply; used in velometers (velocity pickups); output can be integrated to get displacement or differentiated to get acceleration.
- Advantages: robust, low output impedance (little cable problem), high output, no amplifier needed. Limits: bulky, heavy (loads light structures), limited frequency

range (≈ 10 Hz - 1 kHz), sensitive to stray magnetic fields.

- Reverse action - current in the coil in a field produces force $F = B\,l\,i$ → basis of the electrodynamic shaker (Section 5).

Example: $B = 0.8$ T, $l = 5$ m of wire, $v = 0.1$ m/s → $e = 0.8 \times 5 \times 0.1 = 0.4$ V.

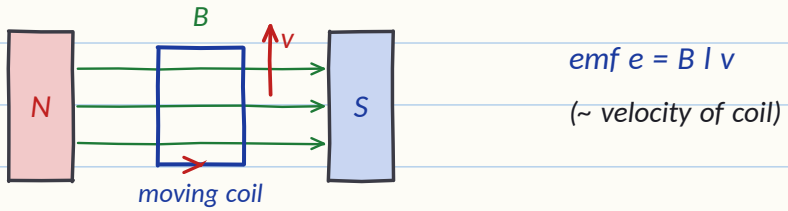


Fig 5.4 Electrodynamic (velocity) transducer

2.4 Linear Variable Differential Transformer (LVDT)

A passive inductive displacement transducer: one primary winding (AC carrier, typically 1-10 kHz) and two identical secondary windings connected in series opposition; a movable ferromagnetic core attached to the vibrating body.

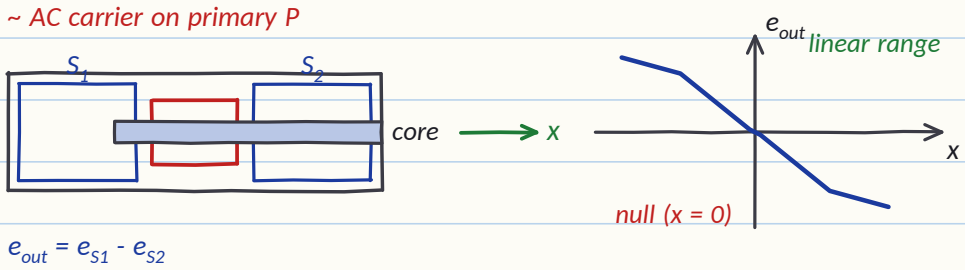


Fig 5.5 LVDT and its output

- Core central ($x = 0$): equal emf induced in both secondaries → net output $e_{out} = 0$ (null).
- Core moves by x: one secondary gets more flux → $e_{out} \sim x$; phase (relative to carrier) tells direction. A phase-sensitive demodulator + low-pass filter gives a DC-level voltage following the motion.
- Carrier frequency ≥ 10 × highest vibration frequency to be measured.
- Advantages: frictionless, infinite resolution, linear over ± few mm to ± 25 cm, robust, good repeatability. Limits: needs AC supply and demodulator, core mass loads small structures, limited bandwidth (carrier).

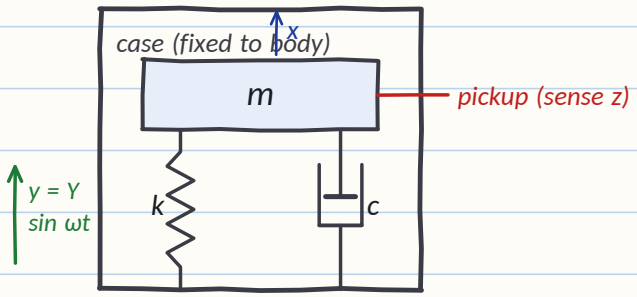
Problem 3 (LVDT)

LVDT sensitivity 100 mV/mm (after demodulation). Output amplitude 0.35 V at 20 Hz. Find the displacement amplitude and peak velocity.

$X = 0.35/0.1 = 3.5$ mm; $v = \omega X = (2\pi \times 20)(3.5 \times 10^{-3}) = 0.44$ m/s.

3. Vibration Pickups (Seismic Instruments)

Most vibration pickups use a seismic mass m on a spring k with damper c, enclosed in a case fixed to the vibrating body. The mass stays almost still in space while the case moves; the relative motion $z = x - y$ between mass (x) and case ($y = Y \sin \omega t$) is sensed (by LVDT, coil, strain gauge, piezo...).



$m\ddot{z} + c\dot{z} + kz = m\omega^2 Y \sin \omega t$

$z = Z \sin(\omega t - \psi)$

$Z/Y = r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$

$\psi = \tan^{-1}[2\zeta r/(1 - r^2)]$, $r = \omega/\omega_n$

$Z/(\omega^2 Y) = (1/\omega_n^2) \cdot 1/\sqrt{[\ldots]}$

Fig 5.6 Seismic pickup and its governing equations

(Equation from Unit II, base excitation, relative motion.) The three types differ only in the range of $r = \omega/\omega_n$ where they are used.

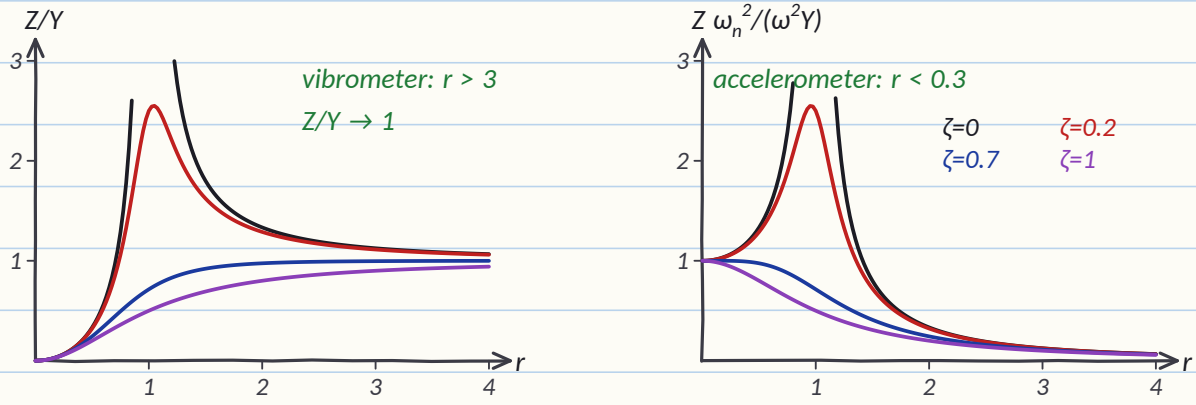


Fig 5.7 Displacement-type (left) and acceleration-type (right) response

3.1 Vibrometer (seismometer - displacement pickup)

Used for $r = \omega/\omega_n >> 1$ ($\omega >> \omega_n$). Then $Z/Y \to r^2/r^2 = 1$, $\psi \to 180°$: the mass remains stationary, relative displacement $z \approx Y$.

- Design: very low natural frequency (soft spring, large mass); useful for $\omega > \sim 3\,\omega_n$ (error small).
- Frequency range: from $\approx 3\omega_n$ upward (no upper limit). Damping $\zeta \approx 0$ - 0.7 ($\zeta = 0$ is ideal for $r > 3$, but $\zeta \approx 0.6$-0.7 gives flat response nearer ω_n).

- Drawbacks: bulky/heavy (soft spring needs large static deflection $\delta_{st} = g/\omega_n^2$; e.g. $f_n = 2$ Hz → $\delta_{st} \approx 62$ mm), cannot be used for small, light structures; large static sag; mounting orientation matters.

Problem 4 (vibrometer error)

Vibrometer $f_n = 4$ Hz, $\zeta = 0$, measures vibration at 40 Hz and reads $Z = 0.5$ mm. Find the true amplitude and error.

$r = 40/4 = 10$; $Z/Y = r^2/|1 - r^2| = 100/99 = 1.0101$ → $Y = 0.5/1.0101 = 0.495$ mm; error in reading = +1.0 %.

3.2 Accelerometer

Used for $r << 1$. Then $(1 - r^2) \approx 1$, $2\zeta r$ small: $1/\sqrt{[...]} \approx 1$, so

$$Z \approx (\omega^2 Y)/\omega_n^2 = (\text{peak acceleration})/\omega_n^2 \rightarrow Z \sim \text{acceleration} \quad (4)$$

- Design: very high natural frequency (stiff spring, small mass). Frequency range from 0 (DC for strain-gauge/capacitive types) to ≈ 0.2-0.5 ω_n.
- Optimum damping $\zeta \approx 0.65$-0.7: response flat within about ±2.5 % up to $r \approx 0.5$ and phase lag nearly proportional to frequency (little phase distortion).
- Sensitivity - bandwidth trade-off: sensitivity ~ $1/\omega_n^2$. Higher ω_n (wider range) → lower output. Piezoelectric types give high ω_n and good sensitivity.
- Advantages: small, light (little mass loading), very wide range, easy to integrate once/twice to velocity/displacement. Also used for shock, impact, seismic, automotive crash tests.

Problem 5 (accelerometer)

Accelerometer $f_n = 1000$ Hz, $\zeta = 0.7$, measuring 100 Hz vibration of peak acceleration 50 m/s^2. Find error and relative displacement of the seismic mass.

$r = 0.1$; M.F. $= 1/\sqrt{[(1 - 0.01)^2 + (2\times0.7\times0.1)^2]} = 1/\sqrt{(0.9801 + 0.0196)} = 1.0002$ → error ≈ 0.02 %.

$\omega_n = 2\pi(1000) = 6283$ rad/s; $Z = (50/6283^2)(1.0002) = 1.27\times10^{-6}$ m = 1.27 μm (tiny - hence sensitive sensing needed).

3.3 Velometer (velocity pickup)

Seismic instrument with an electrodynamic sensor: a coil (or magnet) attached to the mass moves in the field of a magnet fixed to the case. Output emf ~ relative velocity $\dot{z} = \omega Z$.

- Operated at $r > 1$ (usually $r \geq 3$-4): then $Z \approx Y$, so output $e = B\,l\,\omega Y = B\,l$ (velocity of body) - a true velocity pickup; ω_n low (≈ 5-10 Hz).
- Self-generating, rugged, usable at low frequencies; output can be integrated electronically to give displacement, or differentiated for acceleration. Typical use: machinery vibration severity (velocity rms, mm/s).
- Above ≈ 1 kHz coil inductance/resonances limit the response; heavy, sensitive to magnetic fields, orientation dependent.

Pickup	Working range	Output ~	Design requirement
Vibrometer (seismometer)	$\omega >> \omega_n$ ($r > 3$)	displacement	very low ω_n, heavy, soft spring
Velometer	$\omega > \omega_n$ ($r > 1$-3)	velocity	low ω_n, electrodynamic sensor
Accelerometer	$\omega << \omega_n$ ($r < 0.3$-0.5)	acceleration	very high ω_n, light, $\zeta \approx 0.7$

3.4 Phase distortion

Real vibration is usually non-harmonic (periodic): $y(t) = Y_1 \sin \omega t + Y_2 \sin 2\omega t + Y_3 \sin 3\omega t + \ldots$ The instrument shifts each harmonic by its own phase lag $\psi_n = \psi(n\omega)$:

$$z(t) = \Sigma\, Z_n \sin(\, n\omega t - \psi_n \,) \text{ with } Z_n = |G(n\omega)|\; Y_n \quad (5)$$

- Amplitude distortion: M.F. not constant over all harmonics → harmonics amplified differently.
- Phase distortion: if ψ_n is not proportional to frequency, the harmonics are shifted by different time delays $t_n = \psi_n/(n\omega)$ → output waveform differs in shape from the input.

No distortion needs $\psi_n = n\,\omega\,t_0$ (linear phase), i.e. all components delayed by the same time t_0 (6)

- Undamped instrument ($\zeta = 0$): $\psi = 0$ for $r < 1$ and 180° for $r > 1$ - phase does not vary with frequency within each range: waveform preserved (a vibrometer with $\zeta = 0$ and all harmonics above $3\omega_n$ reproduces shape).
- With damping $\zeta \approx 0.7$: for small r, $\psi \approx 2\zeta r$ (linear!), so $t_0 = \psi/\omega = 2\zeta/\omega_n$ = constant ($\approx 1.4/\omega_n$) → minimal phase distortion over $0 < r < 1$. This is why accelerometers are built with $\zeta \approx 0.7$.
- Instruments with $\zeta \approx 0$ used close to resonance ($r \approx 1$) suffer severe distortion (phase jumps 0 → 180°). Low ζ or $\zeta > 1$ (non-linear ψ) should be avoided.

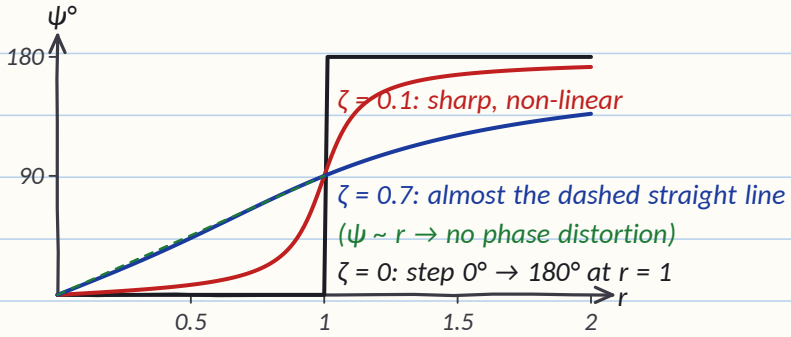


Fig 5.8 Phase lag of a seismic instrument

4. Frequency-Measuring Instruments

(a) Stroboscope

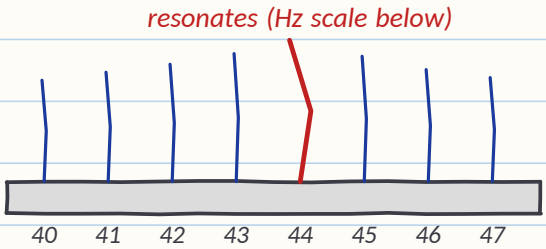
A lamp that flashes at an adjustable, calibrated rate. When flash frequency equals the vibration (or rotation) frequency, the moving part appears stationary (seen at the same position every cycle).

- Procedure: increase the flash rate from low value; the highest flash rate giving a single stationary image equals the frequency; at 2×, 3× ... that rate, one image seen; at flash = f/2, f/3 also one image (sub-harmonics) - to avoid error start from high rate and decrease.
- Non-contact, no loading of the machine, range up to ~ 20 000 flashes/min; also used to slow-motion-view vibration (flash slightly different from f).

Example: stationary single image at 1800 flashes/min; at 3600 two images (double). So speed = 1800 rpm = 30 Hz (not 3600).

(b) Vibrating reed frequency meter (Fullarton / Frahm tachometer)

A row of thin steel reeds of different lengths (each tuned to a different natural frequency, e.g. 40, 41, 42 ... Hz) fixed to a base which is pressed against the vibrating machine. The reed whose natural frequency matches the vibration resonates with maximum amplitude; frequency is read from the scale at that reed.



body vibrates at f →
reed with $f_n \approx f$ shows
largest amplitude

Fig 5.9 Vibrating reed tachometer

- Single-reed (Frahm) type: one reed whose length can be altered by a calibrated screw; tune until amplitude is maximum.
- Range ≈ 10-1000 Hz; simple, cheap; accuracy limited by reed spacing (±0.5 Hz).

(c) Oscilloscope / Lissajous figures, counters and analysers

- Lissajous figures: unknown signal on Y-plates and known generator on X-plates of an oscilloscope. Frequency ratio f_y/f_x = (number of horizontal tangents)/(number of vertical tangents) = 1:1 ellipse/circle, 1:2 figure-eight, 2:3 ...

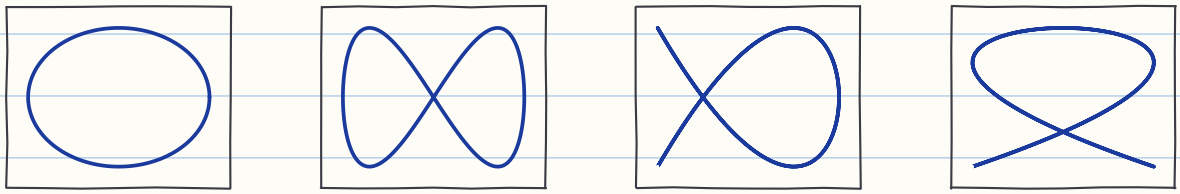


Fig 5.10 Lissajous figures for $f_x : f_y$

- Digital counters count zero-crossings (cycles per second) of conditioned signal; FFT (spectrum) analysers display amplitude vs frequency of all components (real-time, used for machine diagnostics).
- Tunable band-pass filters (octave / third-octave) and tracking filters give amplitude at selected frequency or running-speed orders.

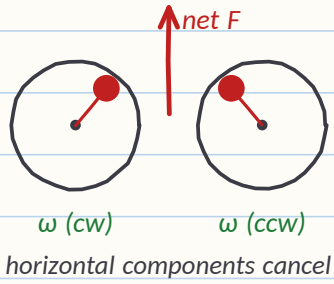
5. Vibration Exciters (Shakers)

Devices that apply controlled vibratory force/motion to a structure or specimen for: resonance testing, modal testing (FRF), fatigue and qualification (environmental) tests, calibration of pickups.

5.1 Mechanical exciters

- *Unbalanced rotating-mass (eccentric) exciter*: eccentric mass m_0, eccentricity e, speed $\omega \rightarrow$ rotating force $m_0 e\omega^2$. For a *unidirectional force* use two *counter-rotating* equal masses, horizontal components cancel:

$$F(t) = 2\, m_0\, e\, \omega^2 \sin \omega t \quad \text{(force amplitude} \sim \omega^2\text{)} \tag{7}$$



Eccentric masses on geared shafts, speed variable (motor + gears) Force changes with ω^2; useful range ≈ 5-100 Hz, force to ~ 20 kN

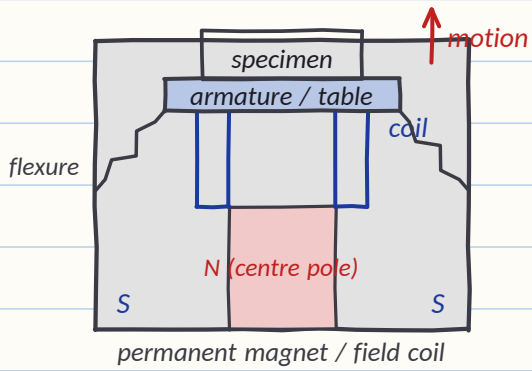
Fig 5.11 Counter-rotating mass exciter

- *Advantages:* simple, cheap, large forces at low frequency, easily mounted on structures (buildings, bridges, machine foundations). *Limits:* force depends on speed (cannot hold force constant), poor at very low frequency (force ~ ω^2), frequency range narrow, no waveform control (sinusoidal only).
- *Scotch-yoke / crank-slider / cam exciters*: a motor drives a crank, slider or cam which gives *constant displacement amplitude* (stroke) $x = X \sin \omega t$ to the specimen - used for low-frequency, large-displacement tests and fatigue; amplitude set by crank radius, frequency by motor speed (1-60 Hz).
- *Hydraulic (electro-hydraulic) exciters*: servo-valve controlled actuator; large forces and strokes at low frequency (0.1 Hz - 500 Hz) - earthquake simulators, road simulators. Also *impact hammer* and *piezo shakers* for modal testing.

5.2 Electrodynamic shaker

Most widely used for laboratory testing. Works on the *motor principle*: current i through a coil placed in a strong radial magnetic field produces force

$$F = B\, l\, i \quad (B = \text{flux density},\ l = \text{length of conductor in field}) \tag{8}$$



Signal generator → power amplifier → coil

Accelerometer on table → feedback control

Limits: stroke (low f), force (mid f),
armature resonance (high f ≈ 5-10 kHz)

Range ≈ 5 Hz - 10 kHz, force 10 N - 200 kN

Waveforms: sine, random, shock

Fig 5.12 Electrodynamic shaker (section)

- Working: amplifier drives AC current through the moving (armature) coil suspended in the radial gap of a permanent magnet (small) or DC field-coil electromagnet (large); force F = Bli moves the table and specimen at the signal frequency and waveform.
- A control accelerometer on the table gives feedback to hold the programmed acceleration; table supported by flexures, cooled by blown air.
- Advantages: wide frequency range, precise control of amplitude and frequency, any waveform (sine sweep, random, shock), low distortion.
- Limits: small stroke (≈ 25 mm) → poor at low frequency; magnetic stray field; armature resonance limits high frequency; heavy/costly; force falls if specimen heavy.

Problem 6 (electrodynamic shaker)

B = 1.2 T, length of coil in field 12 m, current 5 A; moving table + specimen mass 3 kg. Find force, acceleration, and displacement amplitude at 50 Hz.

$F = Bli = 1.2 \times 12 \times 5 = 72$ N; $a = F/m = 24$ m/s^2 (2.45 g); $X = a/\omega^2 = 24/(314.2)^2 = 0.243$ mm.

Problem 7 (counter-rotating exciter)

Two masses of 0.2 kg at e = 0.05 m, rotating at 20 Hz in opposite directions. Find force amplitude.

$\omega = 2\pi(20) = 125.7$ rad/s; $F = 2\, m_0 e\omega^2 = 2(0.2)(0.05)(15\,791) = 316$ N.

6. Applications of Vibration Measurement

Application	What is measured / how
Condition monitoring	Overall velocity rms (ISO 10816 severity: mm/s) and spectra with accelerometers on bearing housings; trending detects faults early
Fault diagnosis (spectrum)	Unbalance: 1× rpm, radial. Misalignment: 1× & 2× rpm (axial). Looseness: harmonics 1×, 2×, 3×.... Gear fault: tooth-mesh frequency = teeth × rpm with side bands. Rolling bearing: ball-pass frequencies (high freq., envelope analysis).
Modal testing	Impact hammer or shaker input + accelerometer output → FRF; peaks give ω_n, half-power gives ζ, phase gives modes
Field balancing	Vibration amplitude and phase with stroboscope/ vibration analyser; trial weights
Structural health monitoring	Strain gauges / accelerometers on bridges, buildings, aircraft - change in natural frequency indicates damage
Product testing	Shakers: sine-sweep, random, shock to qualify electronics, vehicles, packaging

One-plane field balancing (trial-weight method)

1. Measure original vibration vector O (amplitude and phase) at running speed.
2. Add trial weight W_t at a marked angular position; measure new vector O + T.
3. Effect of trial weight T = (O + T) - O (vector subtraction).
4. Correction weight $W_c = W_t \times (O/T)$ in magnitude, placed at an angle equal to the angle of trial weight, advanced by (angle of O to T) in the proper sense (so that its effect = -O).

Numerical example: O = 4 mm/s at 0°; with W_t = 10 g at 0° reading becomes 6 mm/s at 0° → T = 2 mm/s (collinear). W_c = 10 × 4/2 = 20 g placed 180° from the trial position.

7. Quick Revision & Short Questions

Item	Key point
Strain gauge	GF = $(\Delta R/R)/\varepsilon \approx 2$; quarter bridge $e_0 = (E/4)GF\,\varepsilon$
Piezoelectric	$q = dF$; $e_0 = q/C$; self-generating; no DC response; wide range

Electrodynamic	$e = Blv$ (velocity); reverse: $F = Bli$ (shaker)
LVDT	Differential secondaries; $e \sim x$; phase gives direction; carrier ≥ 10 f
Vibrometer	$r >> 1$; $Z \to Y$; low ω_n; $\psi \to 180°$
Accelerometer	$r << 1$; $Z = a/\omega_n^2$; high ω_n; $\zeta \approx 0.7$
Velometer	$r > 1$; electrodynamic; output ~ velocity
Phase distortion	avoid by linear $\psi \sim \omega$; $\zeta \approx 0.7$ for small r
Frequency meters	Stroboscope, reed tachometer, Lissajous, FFT, counters
Exciters	Mechanical: $F = 2m_0e\omega^2$; electrodynamic: $F = Bli$

1. What is a transducer? A device converting mechanical vibration into a proportional electrical signal.
2. Why is a piezo accelerometer popular? High natural frequency, wide range, small, self-generating, good linearity.
3. Why can't piezo sensors measure static acceleration? The generated charge leaks away through finite insulation/input resistance.
4. Why is the LVDT output zero at the central core position? Equal and opposite emfs in the two series-opposed secondaries.
5. Difference between vibrometer and accelerometer? Vibrometer: $\omega >> \omega_n$, low ω_n, measures displacement. Accelerometer: $\omega << \omega_n$, high ω_n, measures acceleration.
6. What is phase distortion? Unequal time delays of harmonic components caused by non-linear phase-frequency response, changing the waveform shape.
7. Why is $\zeta \approx 0.7$ chosen? Widest flat amplitude response and nearly linear phase.
8. Principle of electrodynamic shaker? Force on a current-carrying coil in a magnetic field, $F = Bli$.
9. Why use counter-rotating masses? To cancel the transverse component and get a pure unidirectional force $2m_0e\omega^2$.

Important questions: (1) piezoelectric and LVDT transducers with sketches (2) theory of seismic instruments: vibrometer, accelerometer, velometer (3) phase distortion (4) frequency-measuring instruments - stroboscope, reed tachometer (5) mechanical exciter and electrodynamic shaker.